H2: Watering indicator for houseplants

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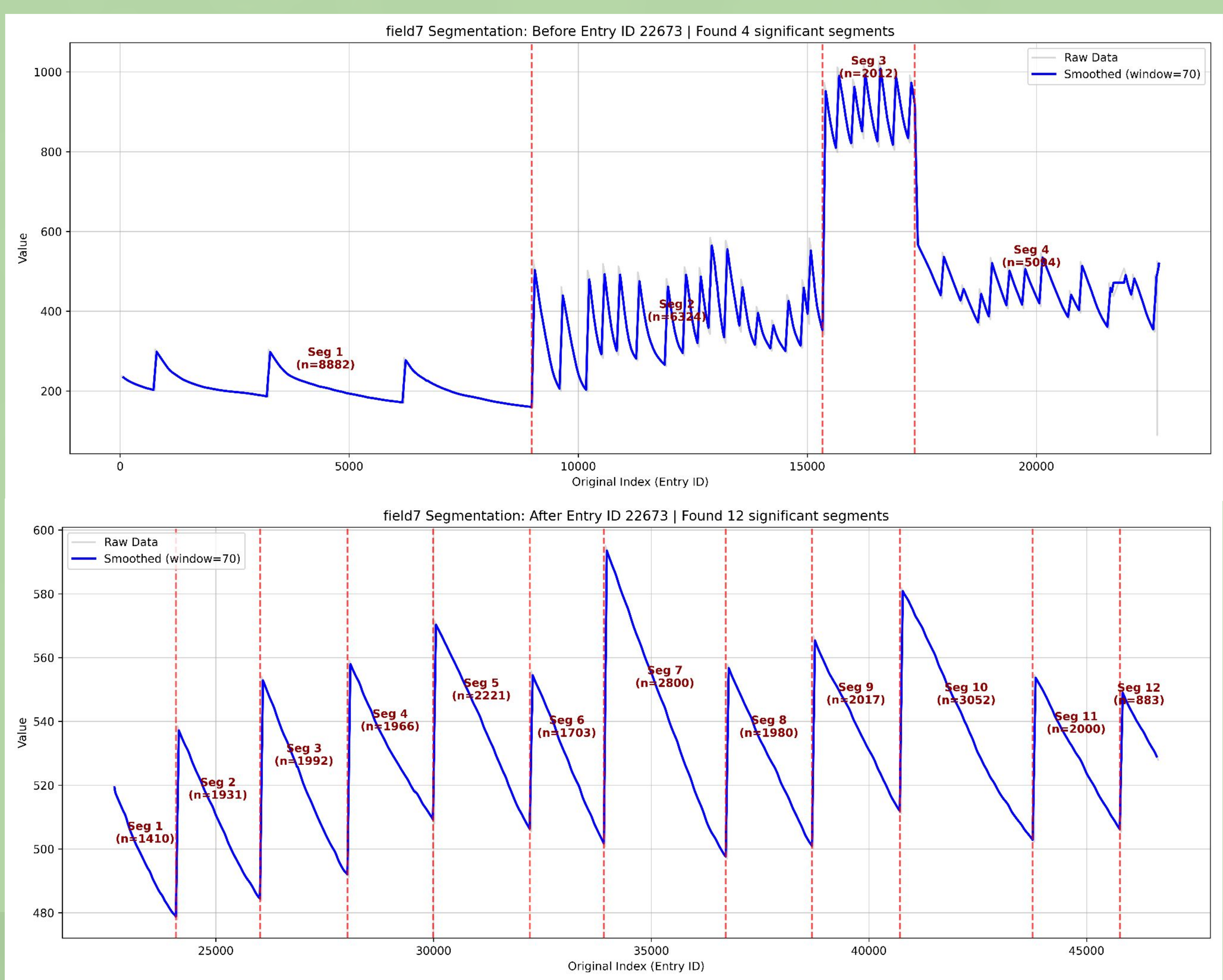
https://github.com/OkiEmil/watering_indicator

Project description

Our project was focused on analysing time series data of plants' weights (pot+plant+soil+water+whatever), to predict when they should be watered. We can determine this, since water loss/intake for plants takes place in **two phases**: Phase 1 is **faster**, **surface-level drying**, and Phase 2 is **slower**, indicating the **critical watering threshold** has been reached. Our coordinator was the company Click & Grow, who have trained a model of their own, and were looking for people to look at the problem with a fresh mindset. It is important to state that this model could never be 100% accurate, since there are so many variables at play.

Data

- The dataset comprised **46,647 time-series datapoints** detailing the weight (in grams) of houseplants across eight separate measurement scales. Data acquisition occurred in two phases:
- An initial **474-day period** (22 673 points) at **≈30-minute intervals**, followed by a high-frequency phase (23 974 points) at **≈5-minute** intervals over **83 days**.
- The high-frequency (5-minute) data exhibited **significantly reduced noise** and **greater measurement stability**.
- The data had multiple problems during its collection including power outages, plants dying, overwatering and other undefined anomalies, that had to be worked around.
- We also tried creating a new "Time-of-day" column, since plants' water intake depends heavily on whether it is day or night



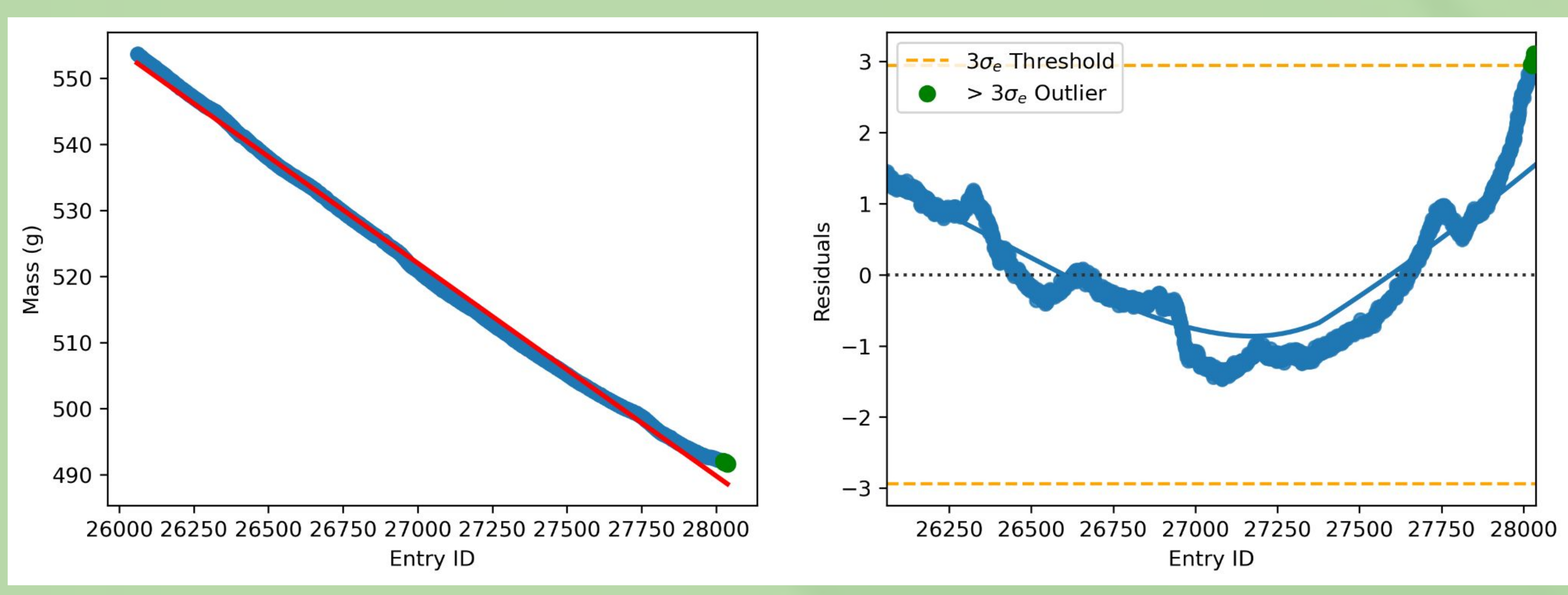
The data series segmented between 30-minute (top) and 5-minute intervals (bottom).

Modeling

The crux of the problem was trying to determine the point, where **phase 1 drying** swapped to **phase 2 drying**. The goal is to predict this point as early as possible.

Methods:

- **Linear regression** in combination with **residual analysis**, since our data is heteroscedastic. We decided to use this approach, since we knew that Click & Grow used a similar approach
- **Cumulative sums** of residuals
- Comparing **standard deviations**
- Comparing **means** of residuals
- LOWESS smoothing to help reduce the noise
- A mathematical approach comparing the **means of absolute water loss** instead of using whole weights



Our replica of Click & Grow's model using three times standard deviation of residuals

Evaluation:

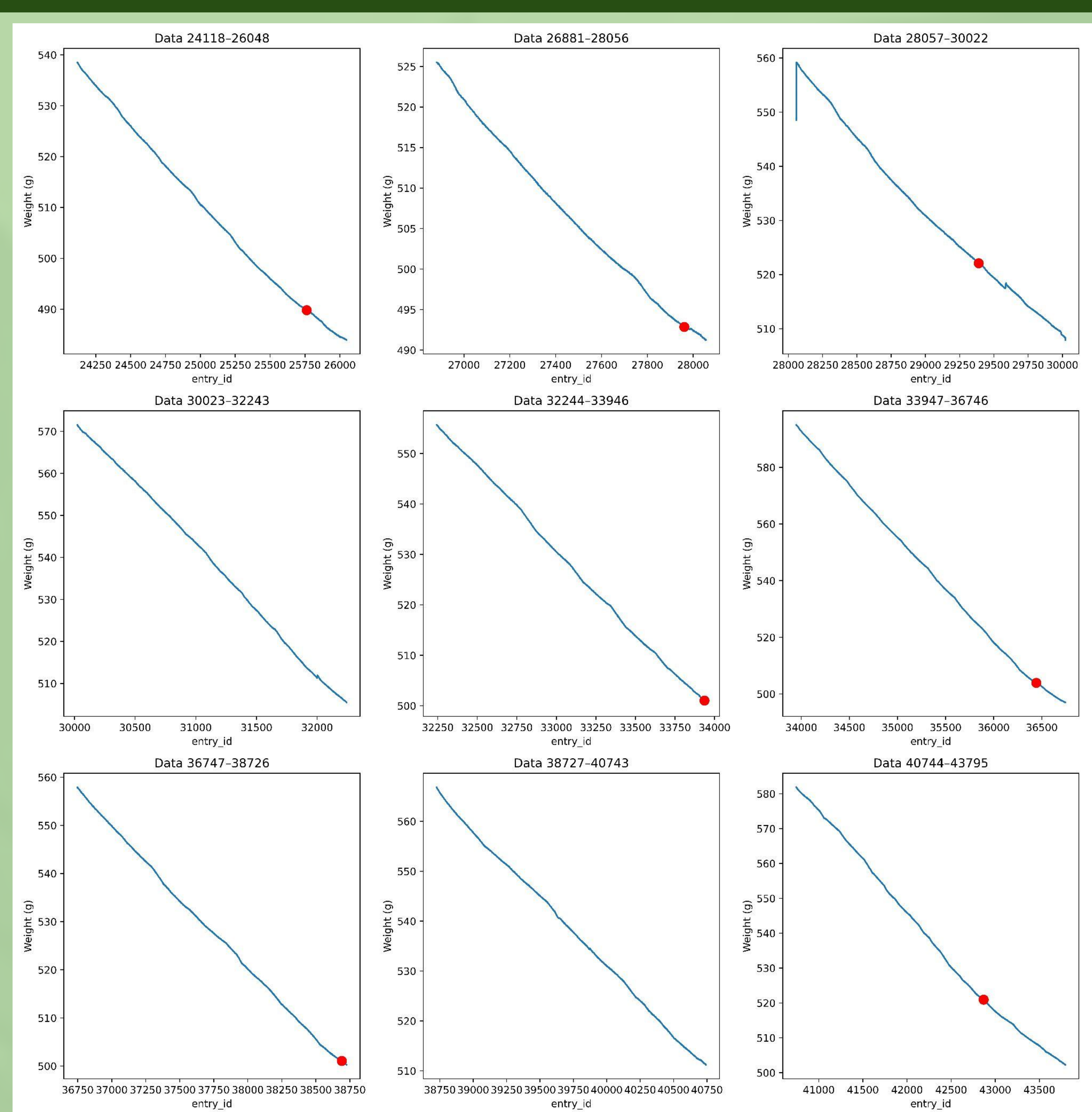
- The dataset did not contain information about the "right time" to water the plant
- The result could be thought of as the start of a range, not a certain point, where you should water the plant
- We can roughly compare our results to the results of Click & Grow's model, since they said that they watered the plant within 3 business days from the prediction
- We tried tweaking the hyperparameters in many different ways, but most of our attempts flunked
- Ideally the mass at the point where the plant needs to be watered does not change much from cycle to cycle

Results

None of our models turned out to be sufficiently accurate on this data, often predicting the phase change either **too early** or sometimes **too late**. Training the models was more complex than initially expected, since plants are quite complex in their structure, and there are many variables in play. Our models were viable, but not as precise as the Click & Grow's model.

Key takeaways for the future:

- Analysing the differences in **absolute water loss instead of total mass** could be really useful, since our models, which were based on this approach, performed the best
- Using **standard deviations** is most likely the best way to explore this type of residual analysis, since Click & Grow's approach showed better results than our alternate methods
- There could be some value in using the **day-night cycle** to predict when to water a plant



Our predictions marked by the red dots